

Section 5.6

I. Differential Equation

- A. Definition $\frac{dy}{dx} = -3y$, $xy' + y = \cos x$,
 $y'' - 3y' + 2y = 0$
- B. Solution $y = 2e^{-3x}$, $y = \frac{\sin x}{x} - \frac{4}{x}$,
 $y = e^{2x} + 4e^x$
- C. General Solution $y = Ae^{-3x}$, $y = \frac{\sin x + c}{x}$
 $y = Ce^{2x} + De^x$
- D. Particular Solution
- E. Initial Value Problem $\frac{dy}{dx} = -3y$, $y(0) = 2$,
 $xy' + y = \cos x$, $y(\frac{\pi}{2}) = 1$
 $y'' - 3y' + 2y = 0$, $y(0) = -2$, $y'(0) = 1$

II. Separable Differential Equations

- A. $\frac{dy}{dx} = \frac{g(x)}{f(y)}$ or $\frac{dy}{dx} = G(x)F(y)$
- B. General solution is obtained by “separating the variables” and integrating
- C. Examples $\frac{dy}{dx} = \frac{xy}{x^2 + 1}$, $xy' = 2y$,
 $x^2 dy + (y + 1)dx = 0$,
 $\frac{dy}{dx} = \frac{x + 1}{xy}$, $y(1) = 1$

III. Modeling: Growth and Decay

A. Equation $y' = ky$ or $\frac{y'}{y} = k$ or $\frac{dy}{dt} = ky(t)$

1. $k > 0 \rightarrow$ growth; $k < 0 \rightarrow$ decay

B. Solution $y = Ae^{kt}$ where $A = y(0)$

C. Examples

1. Radioactive substance has half-life of 428 years. How much of a 20 gram sample will remain after a period of 120 years?
2. A culture of bacteria is 2.2 mg. Two days later the culture measures 2.8 mg. Estimate the culture size at the end of the 3rd day.